Name :- Divyanka Sagolale

Sec :- B

Batch :- B2

practical4.y

practical4.l

```
%{
#include<stdio.h>
#include<stdlib.h>
int yyerror(char*s);
int yylex();
%}

%token NUM

%%
E:E'+'T|T;
T:T'*'F|F;
F:'('E')'|NUM;
%%
int yyerror(char*s){
        printf("Invalid Expression!!");
        exit (0);
}

int main(){
        printf("Enter an Expression: ");
        yyparse();
        printf("Valid Expression");
        return 0;
}
```

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Open

```
%{
#include<"y.tab.h">
%}

%%
[0-9]+ {return NUM;}
[ \t];
\n {return 0;}
.  {return yytext[0];}
%%

int yywrap(){
        return 1;
}
```

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```
sit@sit-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~/cc lab$ yacc -d practical4.y
sit@sit-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~/cc lab$ flex practical4.l
sit@sit-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~/cc lab$ gcc y.tab.c lex.yy.c
sit@sit-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~/cc lab$ ./a.out
Enter an Expression: 2+3*4
Valid Expressionsit@sit-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~/cc lab$ yacc -d practical4.y
sit@sit-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~/cc lab$ flex practical4.l
sit@sit-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~/cc lab$ gcc y.tab.c lex.yy.c
sit@sit-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~/cc lab$ ./a.out
Enter an Expression: 4=5-1
Invalid Expression!!sit@sit-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~/cc lab$
```

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practical4.y

```
%{
#include<stdio.h>
#include<stdlib.h>
int yyerror(char *s);
int yylex();
%}

%token number identifier

%%
E : E '+' T | E '-' T | T;
T : T '*' F | T '/' F | F;
F : '(' E ')' | number | identifier;
%%

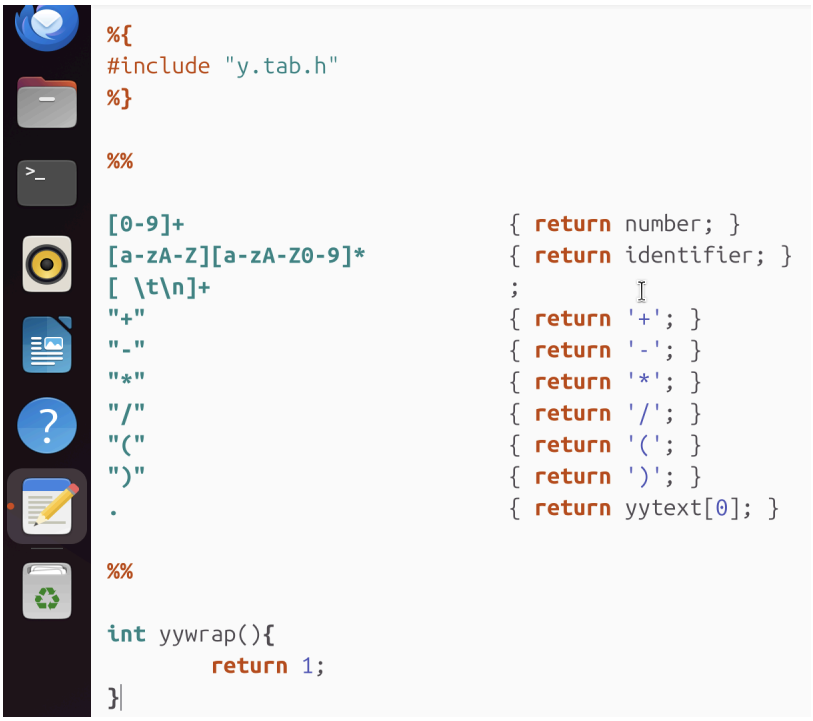
int yyerror(char *s)
{
    printf("Invalid Expression\n");
    exit(0);
}

int main()
{
    printf("Enter an Expression: ");
    yyparse();
    printf("Valid Expression\n");
    return 0;
}
```

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```
%{
#include "y.tab.h"
%}

%%

[0-9]+                          { return number; }
[a-zA-Z][a-zA-Z0-9]*            { return identifier; }
[ \t\n]+                        ;
"+"                             { return '+'; }
"-"                             { return '-'; }
"*"                             { return '*'; }
"/"                             { return '/'; }
"("                             { return '('; }
")"                             { return ')'; }
.                               { return yytext[0]; }

%%

int yywrap(){
        return 1;
}
```

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```
div12@div12:~$ yacc -d practical4.y
div12@div12:~$ flex practical4.l
div12@div12:~$ gcc y.tab.c lex.yy.c
div12@div12:~$ ./a.out
Enter an Expression: a+b*c
Valid Expression
div12@div12:~$ ./a.out
Enter an Expression: a*/b
Invalid Expression
div12@div12:~$
```